

Stuck priors in Qwen3-32B

Proposal-distribution collapse in a restricted word-convergence game — evidence per claim

NOT SUPPORTED

"The model autonomously repeats forever."
Absorbing behavior was handler-dependent: under strict logit-masked novelty every loop escapes in 1–3 turns.

SUPPORTED

"The action distribution becomes heavily concentrated
masked obsolete self-generated policy."
22 resamples/turn mean; 23% of turns exhaust all 64.

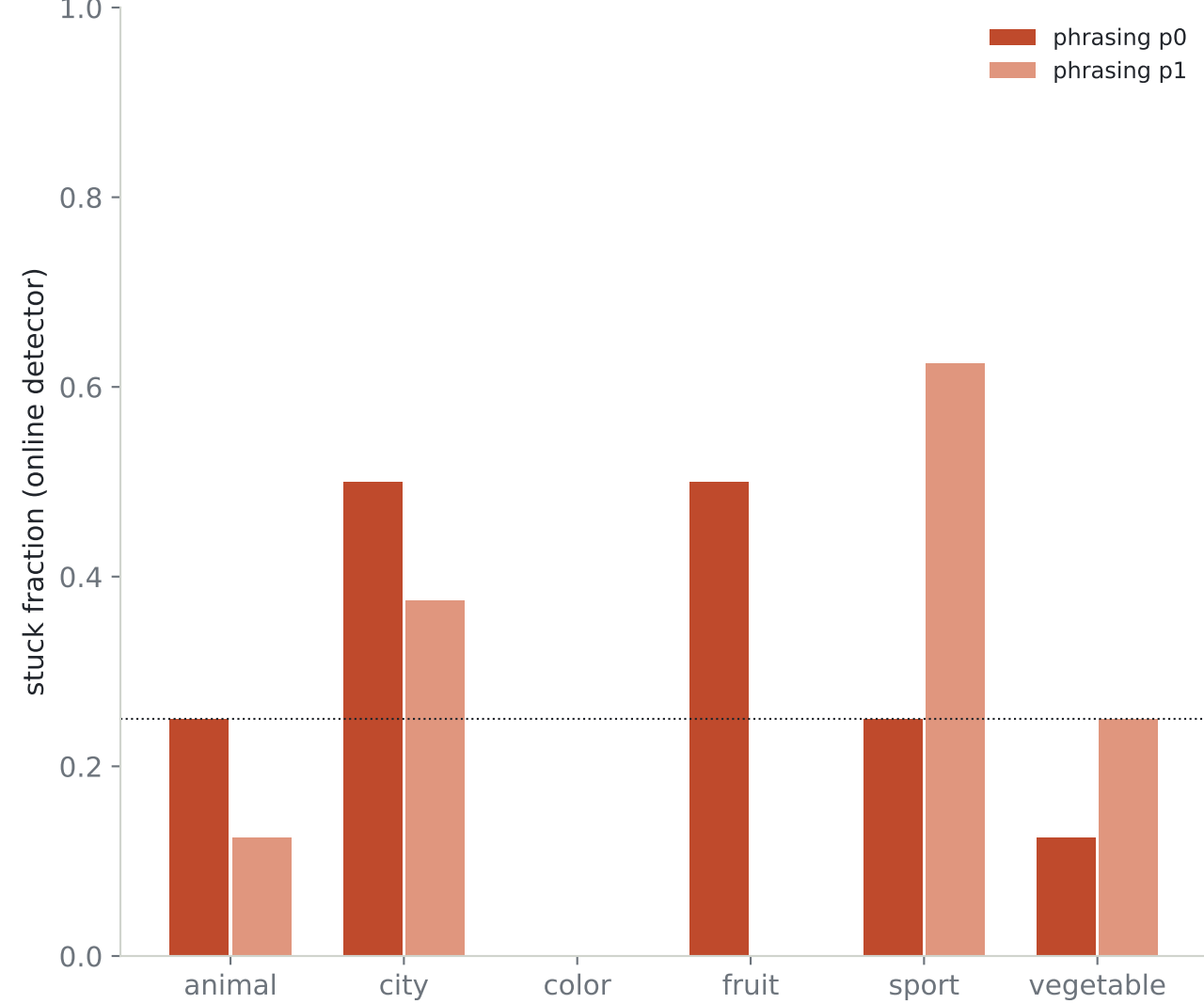
RESOLVED: METASTABLE

The latent attractor persists under forced behavioral novelty —
invalid proposal mass climbs 0.42 → 0.87 onset-aligned and stable
while the handler emits novel words.

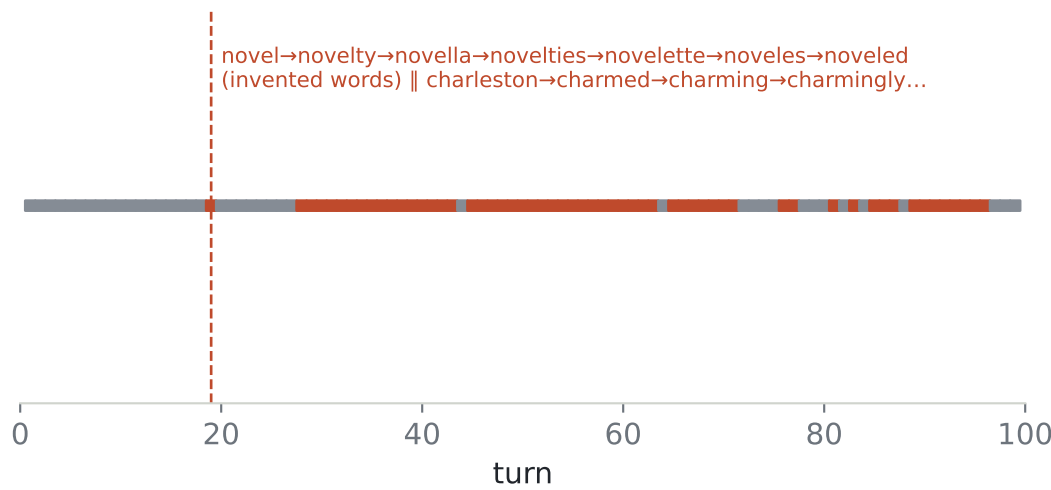
Game: two players say one word per turn, win on saying the same word, no repeats. B is secretly restricted to a category (city, fruit, ...). A (unrestricted) must infer and accommodate. Qwen3-32B as both players; K=64 Monte-Carlo proposal profiling.

A perseverates in self-generated word families instead of adopting B's category

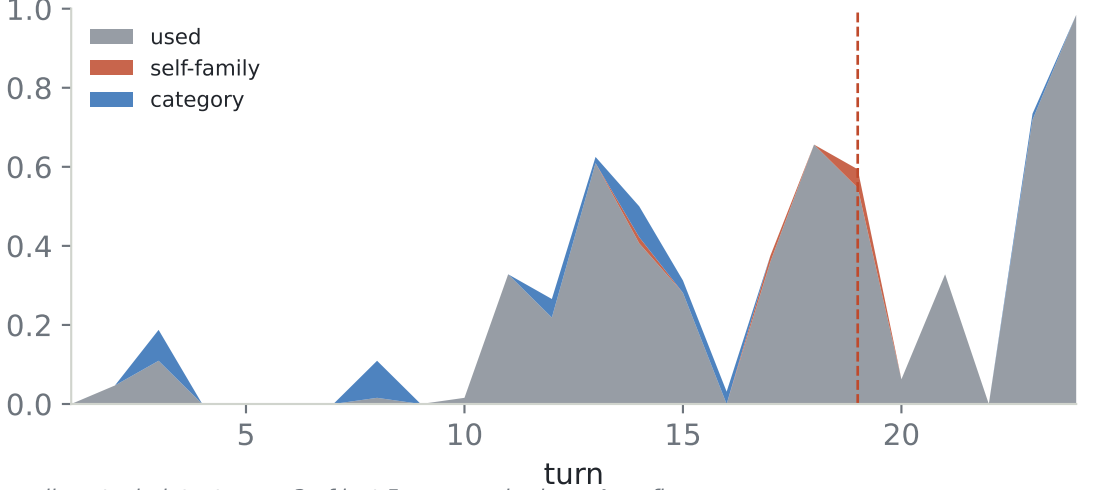
Reproduces across 6 hidden categories × 2 prompt phrasings



Game g0: A's words — family members in rust (onset t19)



g0 proposal mass (K=64 MC), turns 1-24



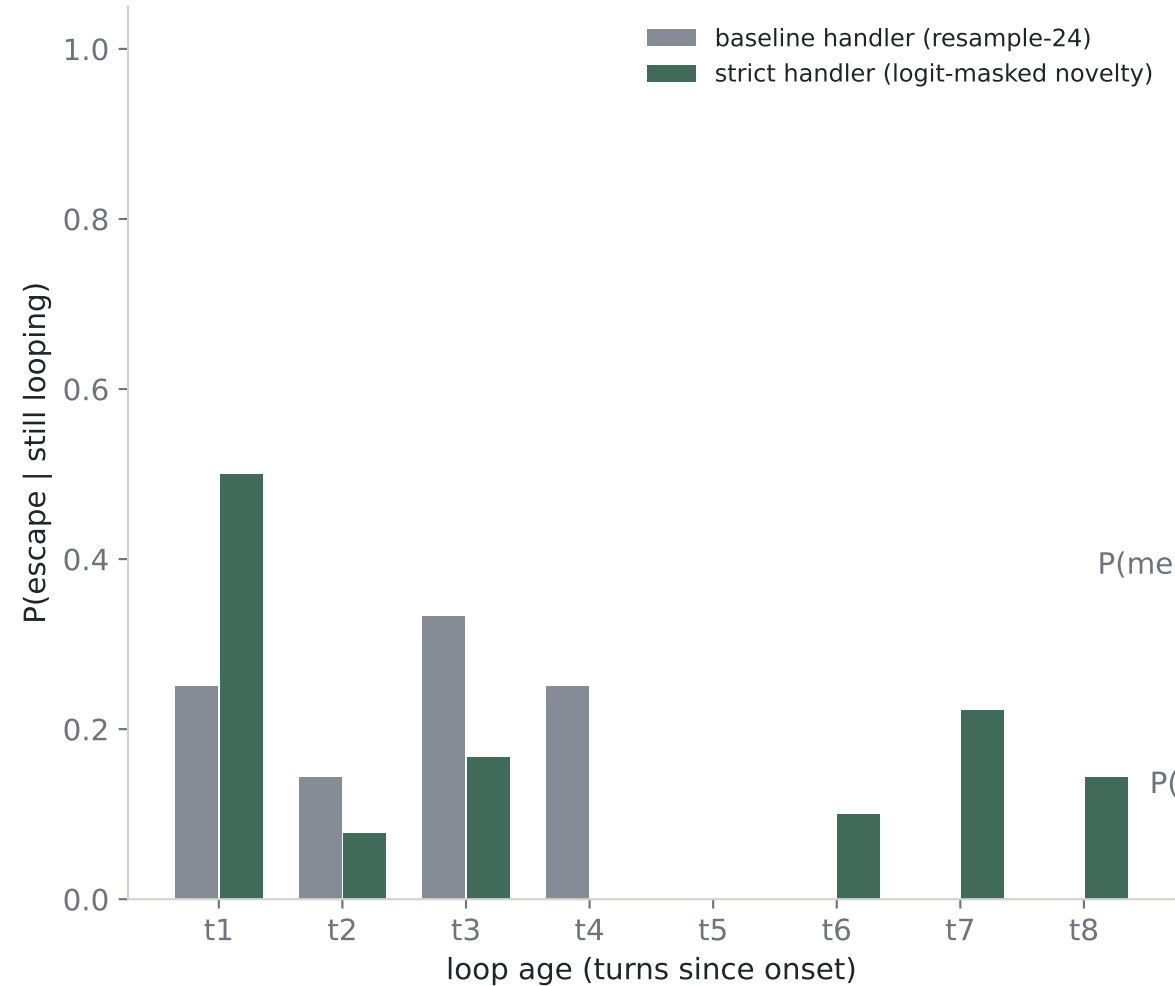
sources: stuck_repro_sweep/stuck_repro_QwenInst32.json · qwen32_strict_cap100/{transcript,proposal_telemetry}.jsonl · online stuck detector = ≥3 of last 5 own words share 4-prefix

"The model repeats forever" — NOT SUPPORTED: trapping was handler-dependent

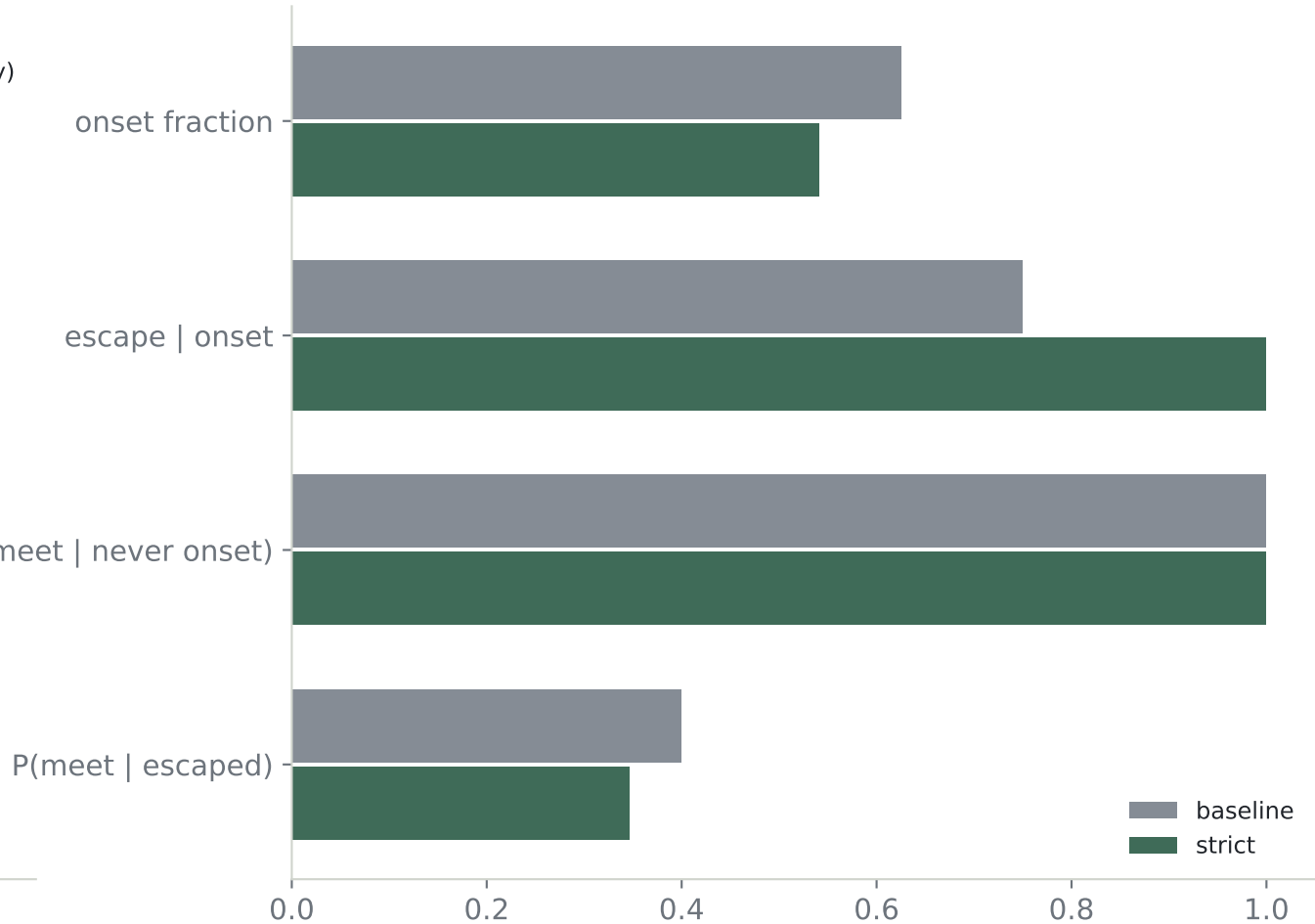
Baseline handler: hazard 0.14–0.33 for loop-turns 1–4, then ≈ 0 — loops became absorbing (durations censored to 197).

Strict handler (guaranteed novelty): EVERY loop escapes (1.00), durations 1–3 turns; onset itself is unchanged — formation is endogenous, trapping was not.

Escape hazard by loop age



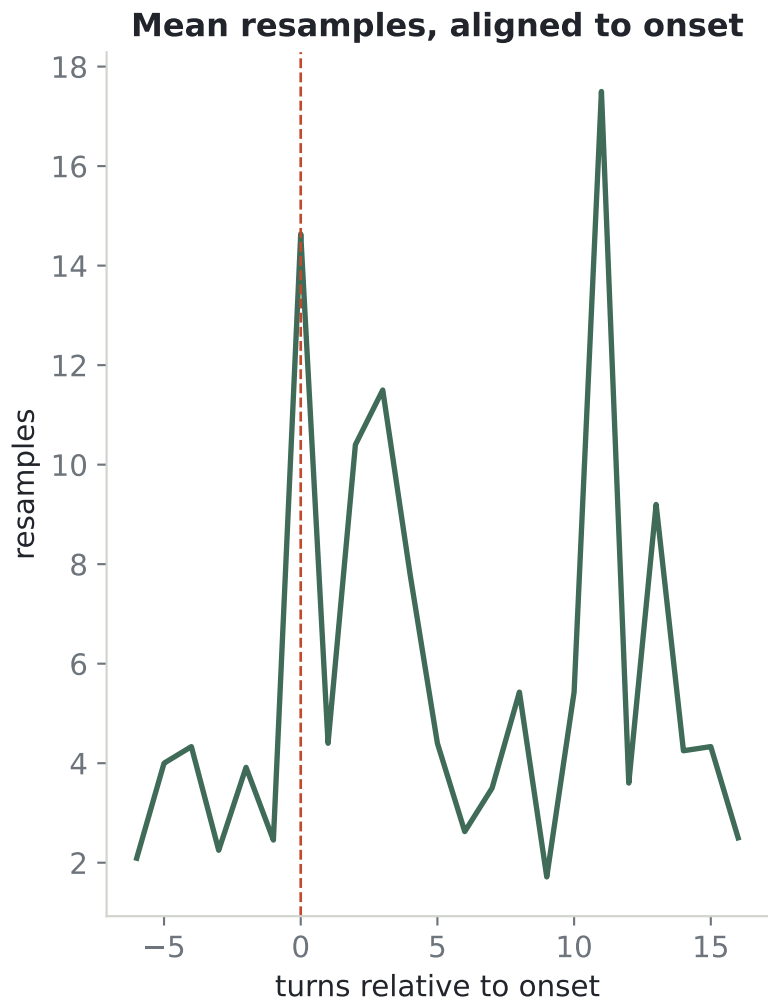
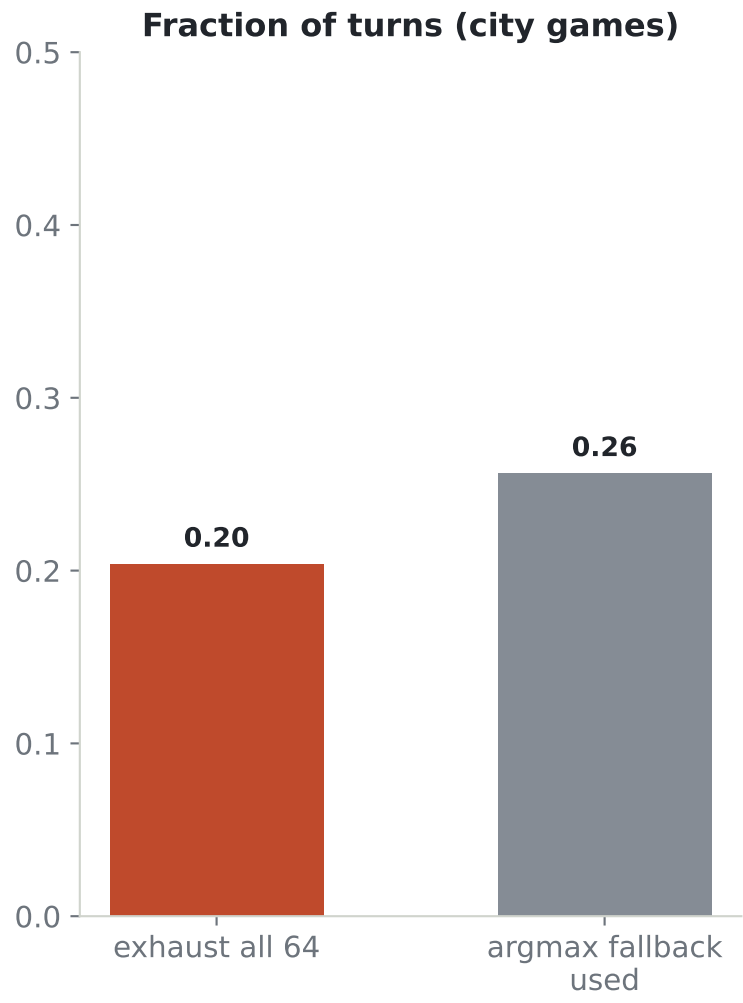
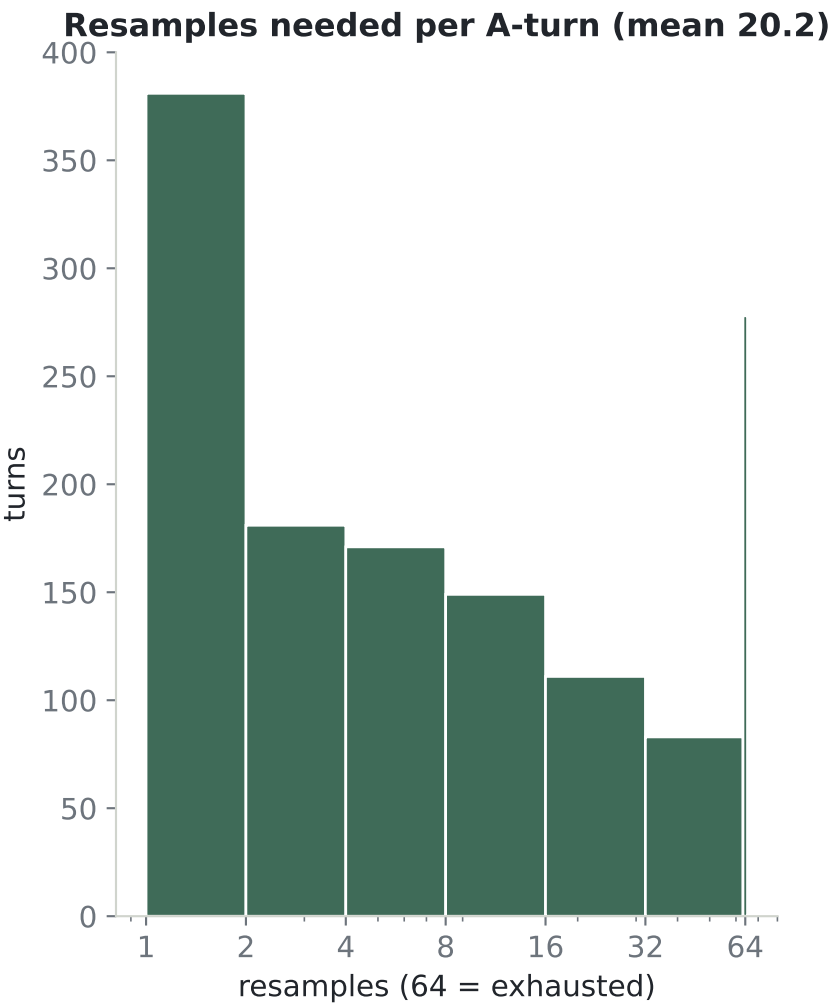
Survival statistics



sources: 6_analyses/stuck_probe/survival.json (n=32, cap-200) and survival_strict.json (cap-100) · onset = first morphological recurrence, escape = 3 consecutive non-family words

The sampling distribution collapses: the handler, not the model, supplies novelty

Under the strict handler A emits novel words every turn — but only because the harness discards a mean of 20 samples/turn; 20% of turns exhaust all 64 and fall back to core. 67% of loop-exit words were delivered by the argmax fallback, not sampled from the model (pod analysis).



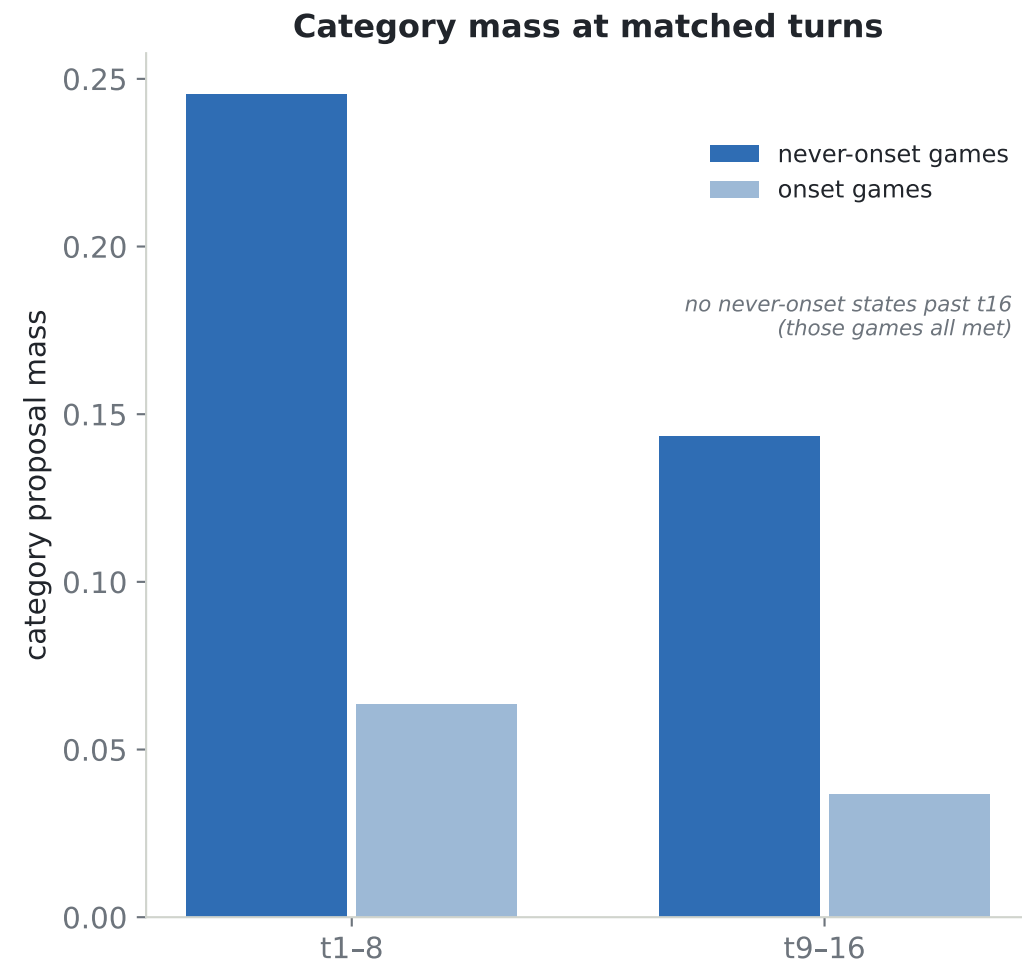
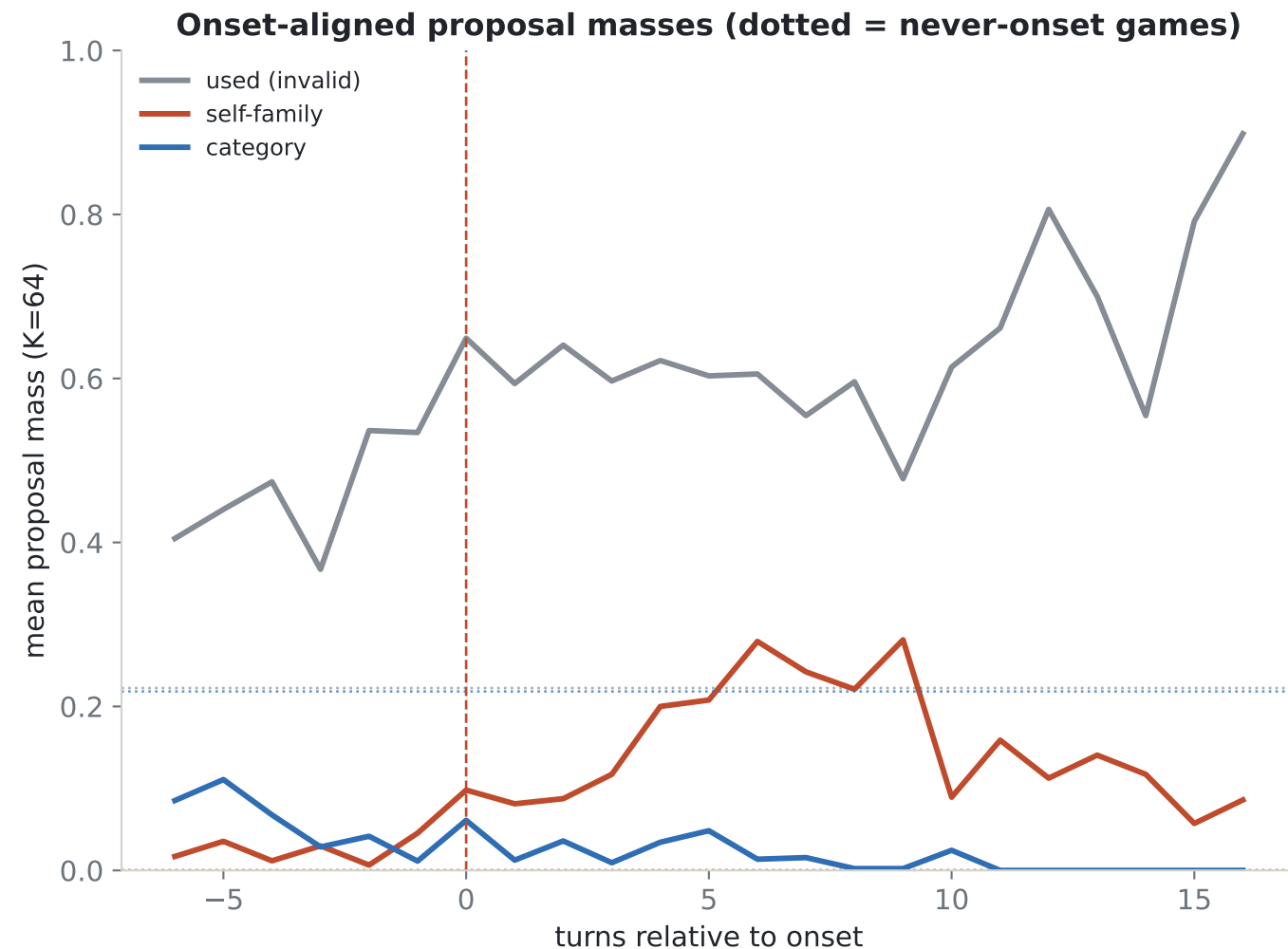
source: qwen32_strict_cap100/game1_strict_city_transcript.jsonl (per-turn resamplesA/fallbackA) · proposal_telemetry.jsonl (onset alignment)

CLAIM 3

The latent attractor is metastable: collapse persists under forced novelty

Invalid (used-word) mass climbs $0.42 \rightarrow 0.87$ across onset and STAYS there while the strict handler forcibly emits novel words each turn.

"Every loop escapes in 1–3 turns" is behaviorally true and mechanistically misleading — the distribution never recovers.

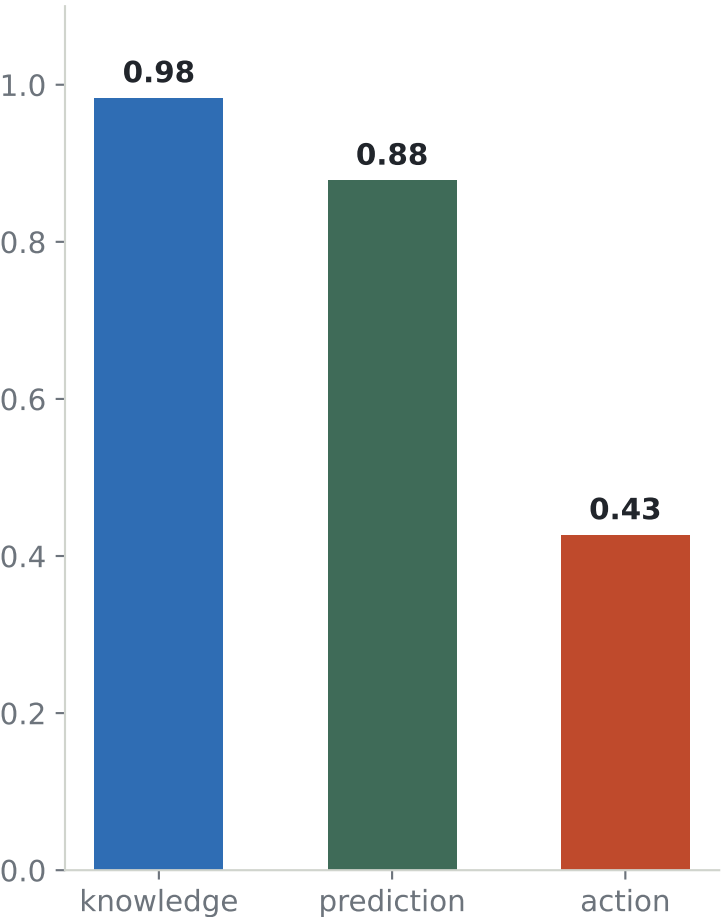


source: proposal_telemetry.jsonl — 725 states $\times$ 64 MC proposals · caveat: used mass partly scales with game length; self-family and category columns are the load-bearing comparisons

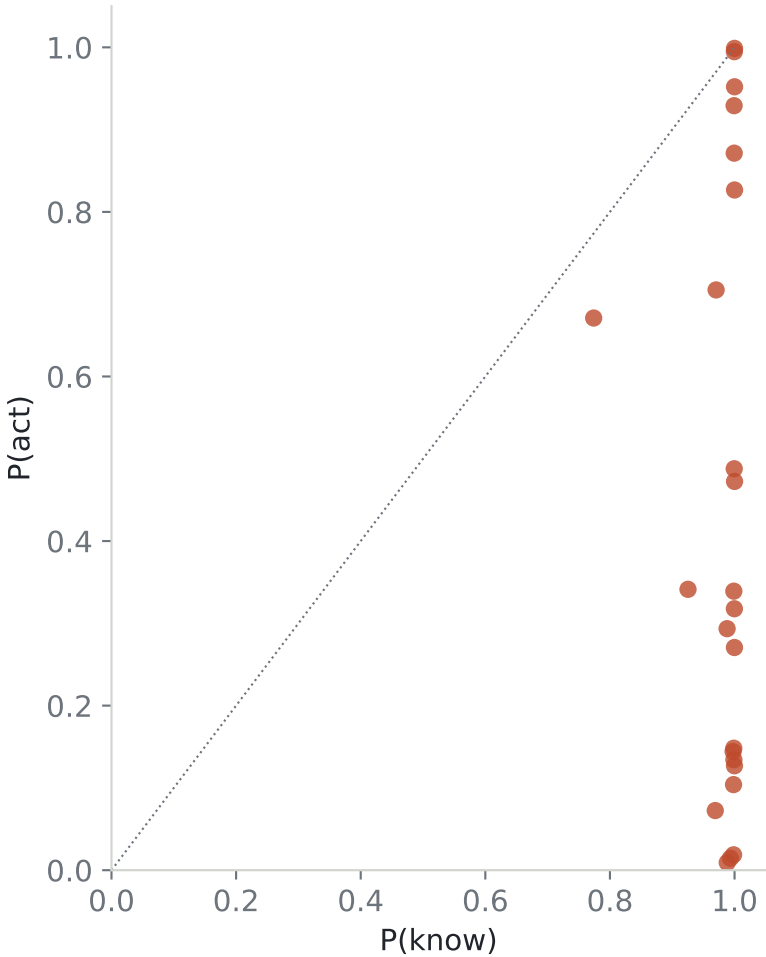
Knowledge is intact; the deficit is specific to the action frame

Same 8 candidates, three question frames, 24 reconstructed loop snapshots: know 0.98 / predict 0.88 / act 0.43 — dissociation +0.56, 95% CI [0.41, 0.69], positive in 24/24.
Forced-choice prediction is CLEAN (0.88) — the earlier 'expects its own loop word' anecdote does not generalize.

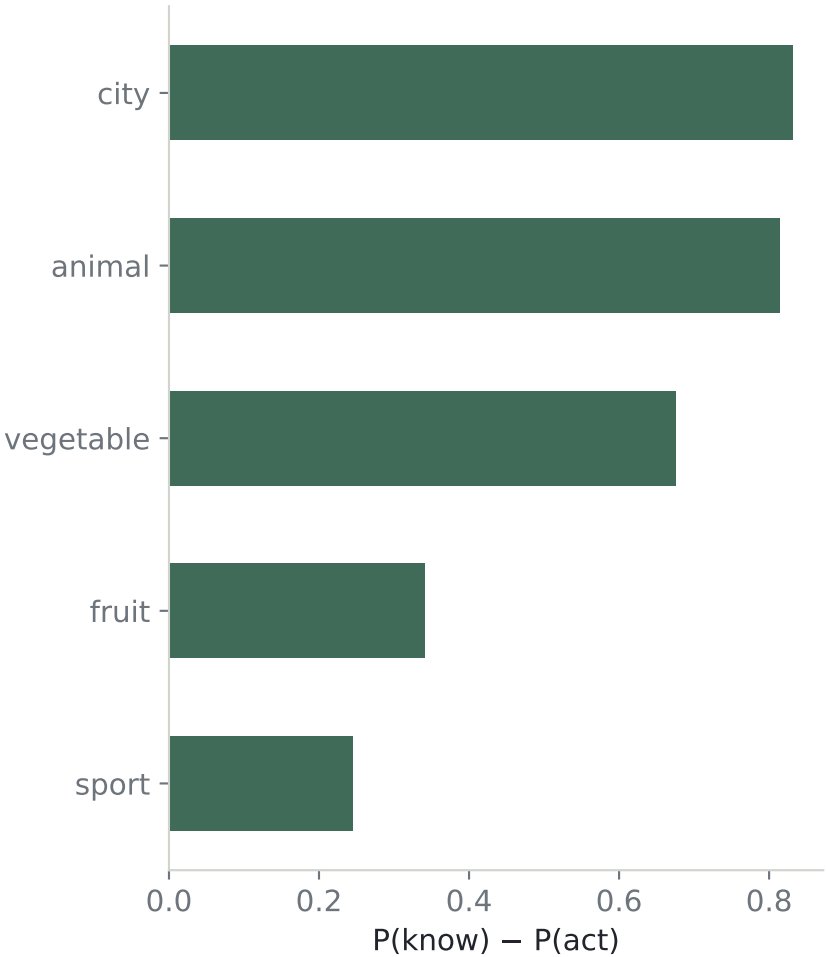
P(correct partner category), forced choice



Per-snapshot (n=24): all below diagonal



Dissociation by category

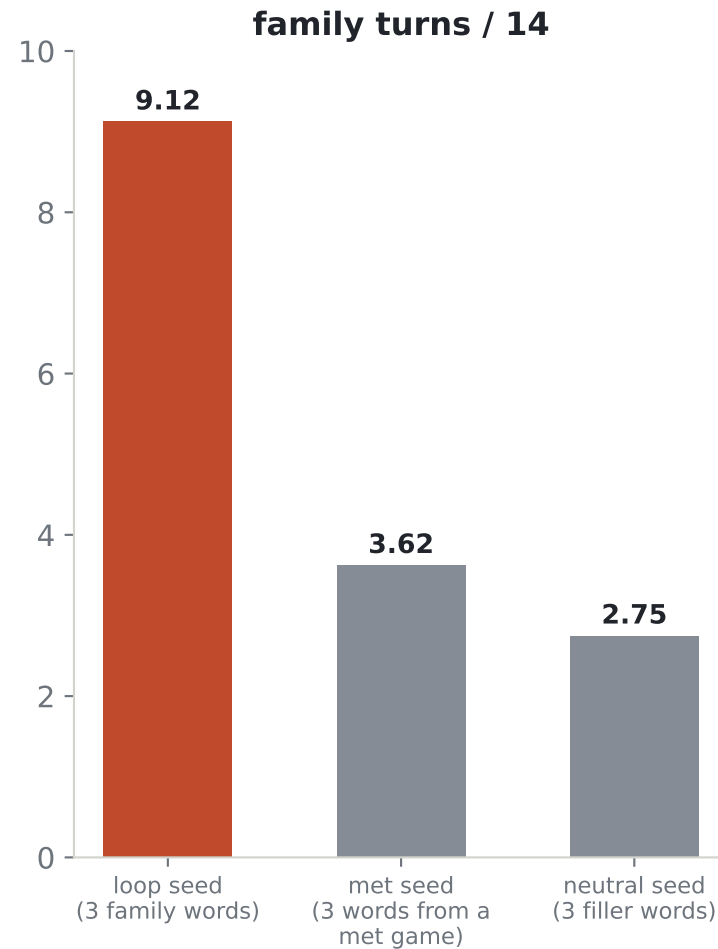
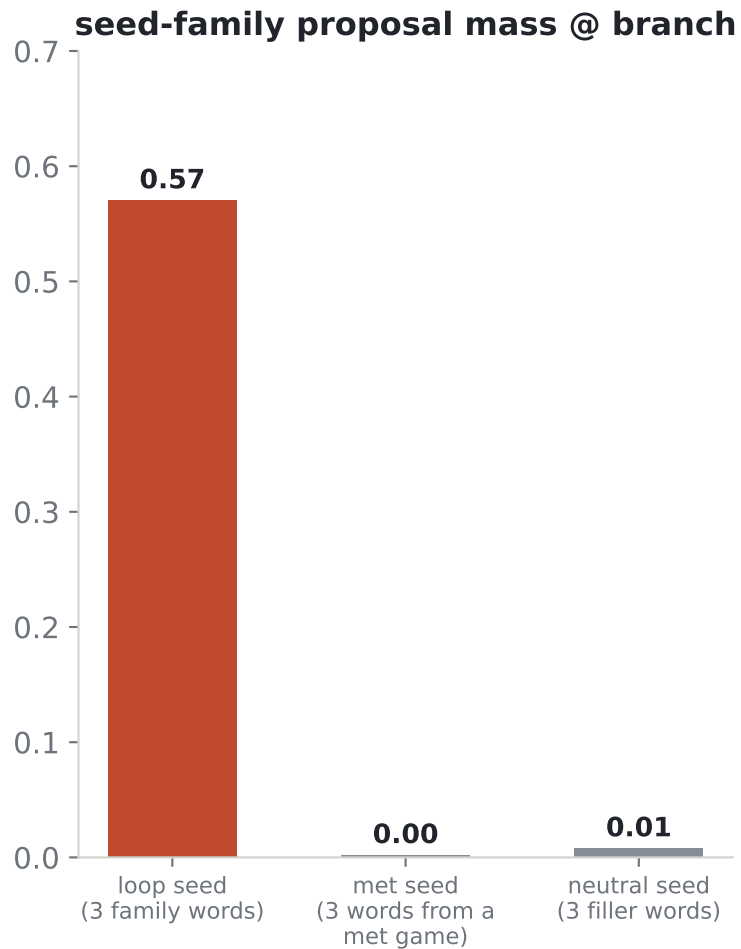
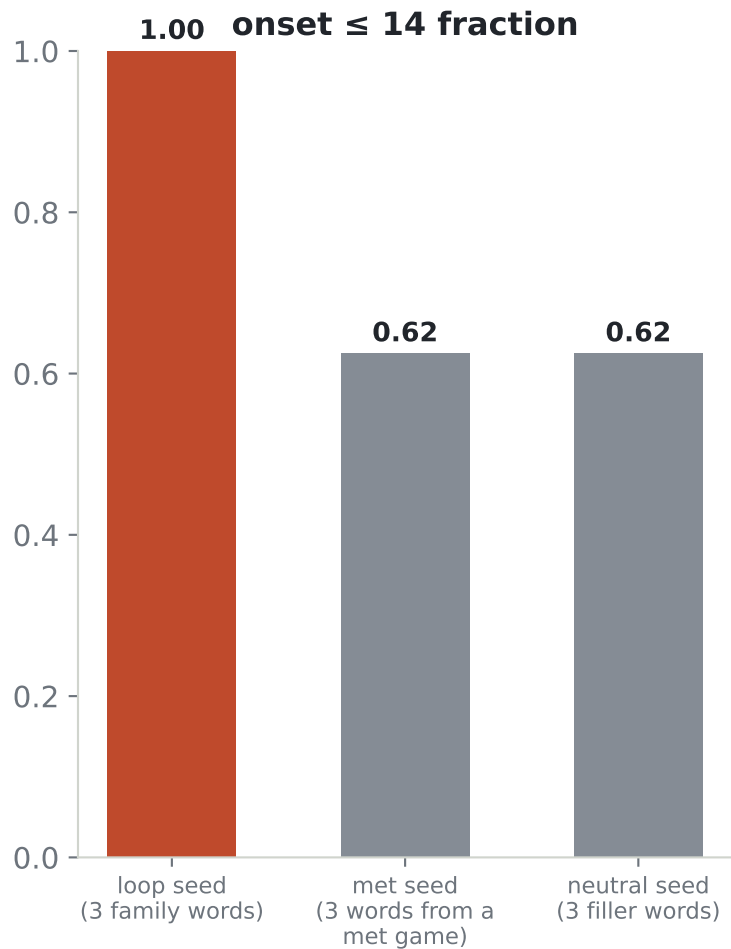


source: 3_interventions/dissoc_branches/dissociation.json · untouched-continuation check: quiz frames do not alter behavior (loop_exit 0.04, met 0.08)

Self-history causally induces the collapse — under identical partner evidence

Branch surgery: replay the SAME partner stream, vary only A's 3 prior own-words (n=8/condition).

Three same-family words instantly move 57% of the proposal distribution into that family and produce onset in 8/8 — controls sit at the ~0.62 base rate with ≈ 0 family mass



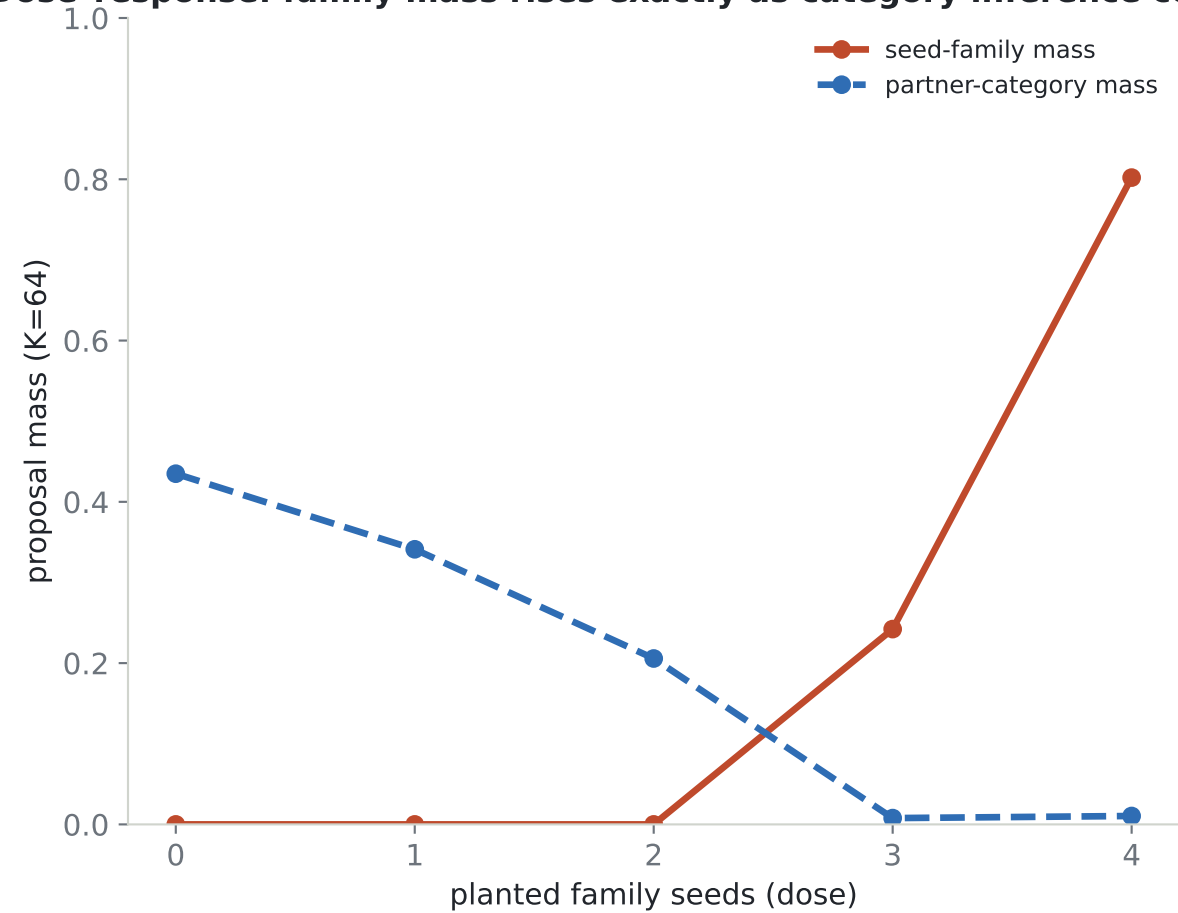
Formation is threshold-gated (3-4 seeds) and hysteretic at dose 4

Morph seeds as A's own words vs identical replayed B streams: mass 0 / 0 / 0 / 0.24 / 0.80 at doses 0-4; category mass 0.43 → 0.01 — the same competition seen in live gan

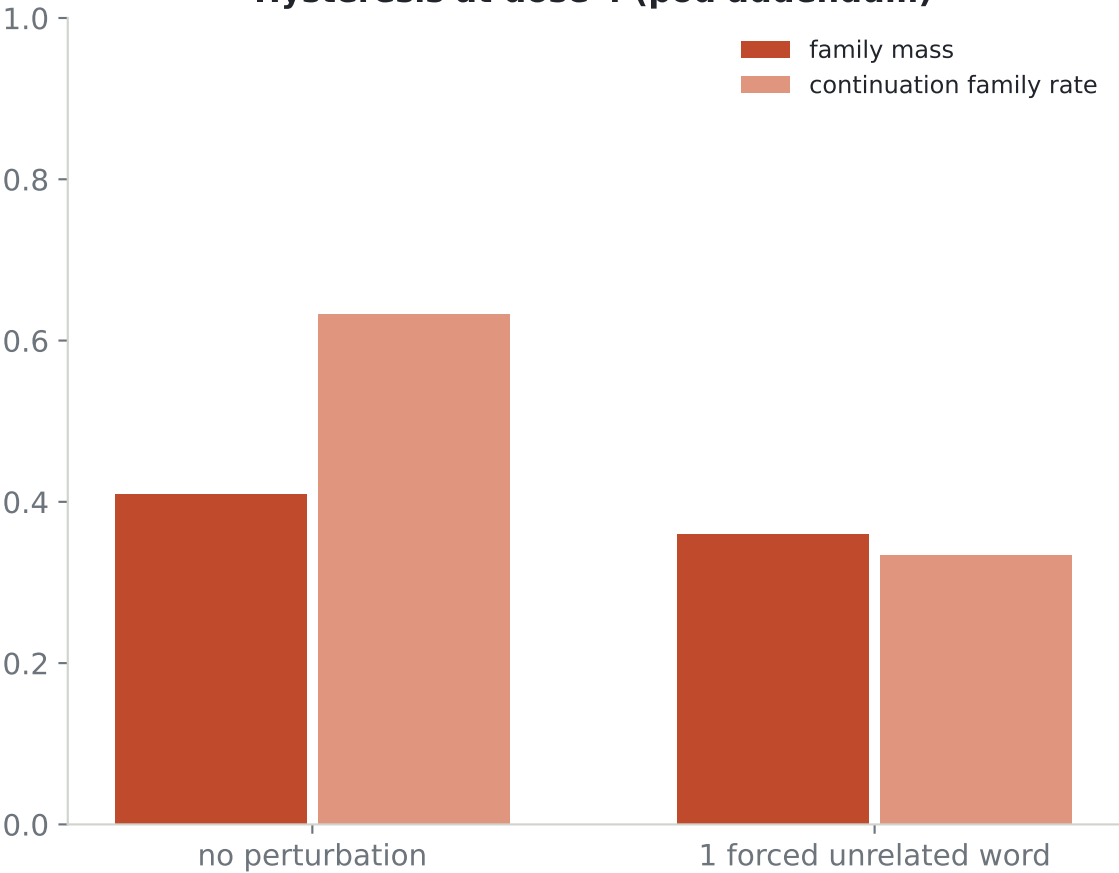
One forced unrelated word barely dents a formed dose-4 attractor (0.41 → 0.36) — persistence, not fragile priming.

Dose-3 perturbation cells were under-dosed (self-decayed) — superseded by the dose-4 addendum shown here.

Dose-response: family mass rises exactly as category inference collapses



Hysteresis at dose 4 (pod addendum)

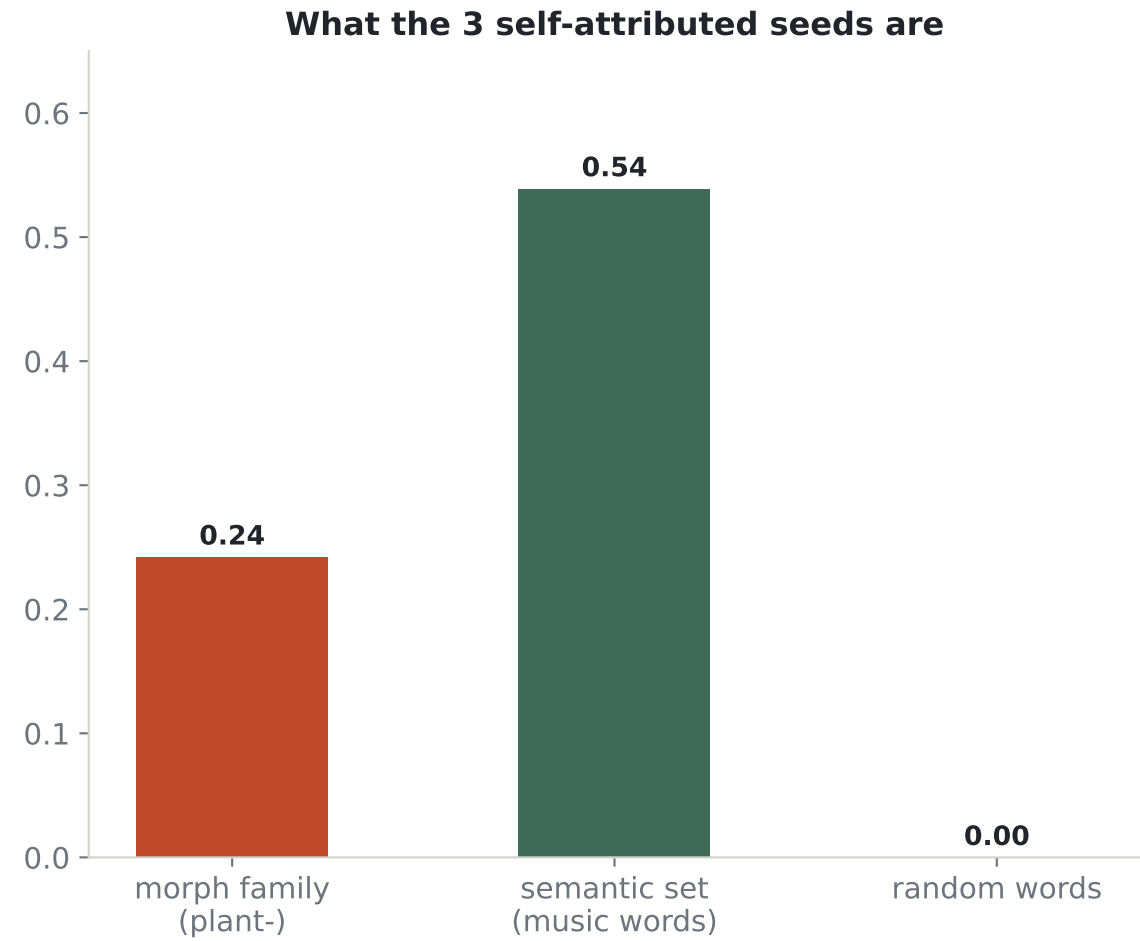
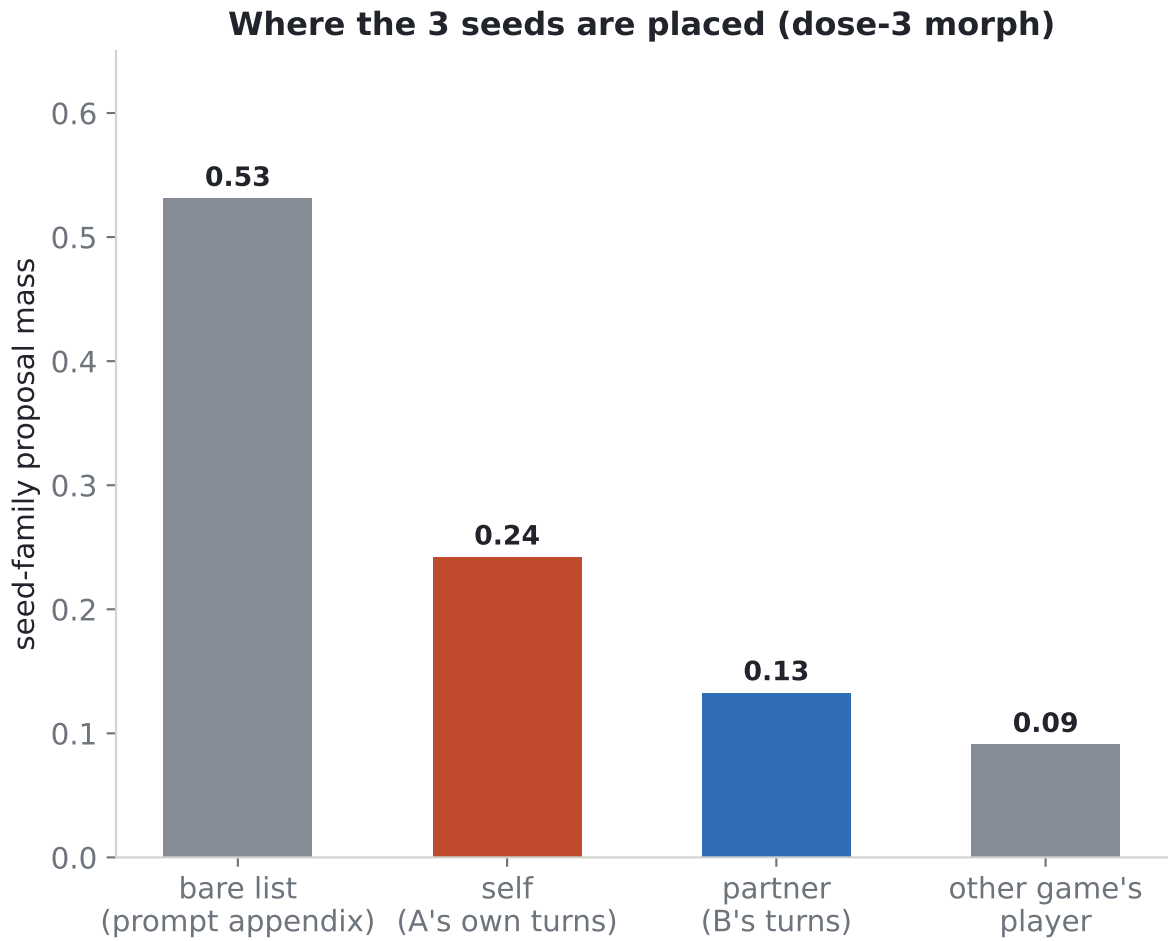


Capture is lexical-contextual: self-attribution is NOT privileged

Bare-list priming (0.53) ≥ self-attribution (0.24) > partner (0.13) > other-game (0.09): lexical presence with attribution-modulated gain — not an ownership effect.

Semantically-related seeds capture mass too (0.54): the mechanism is not morphology-specific.

Known confounds: the partner cell alters partner evidence; list/other differ in position and format from in-history seeds.



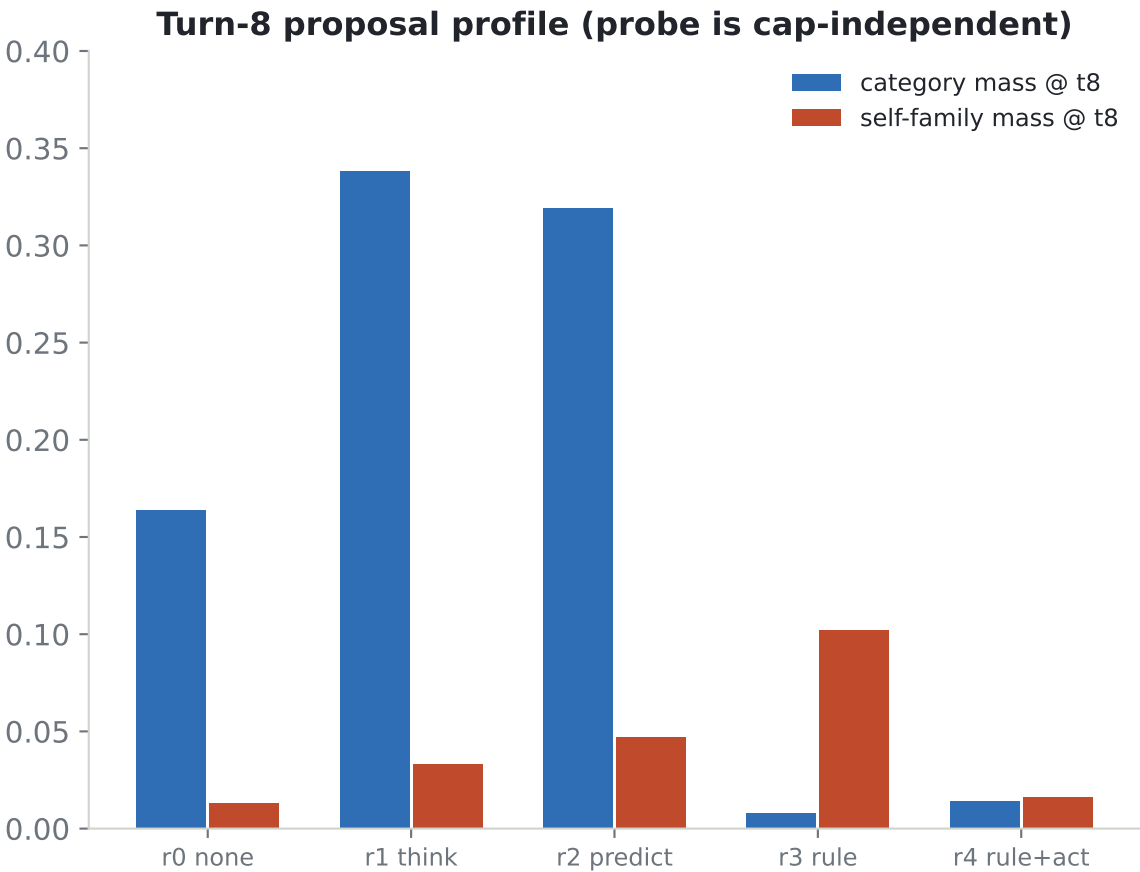
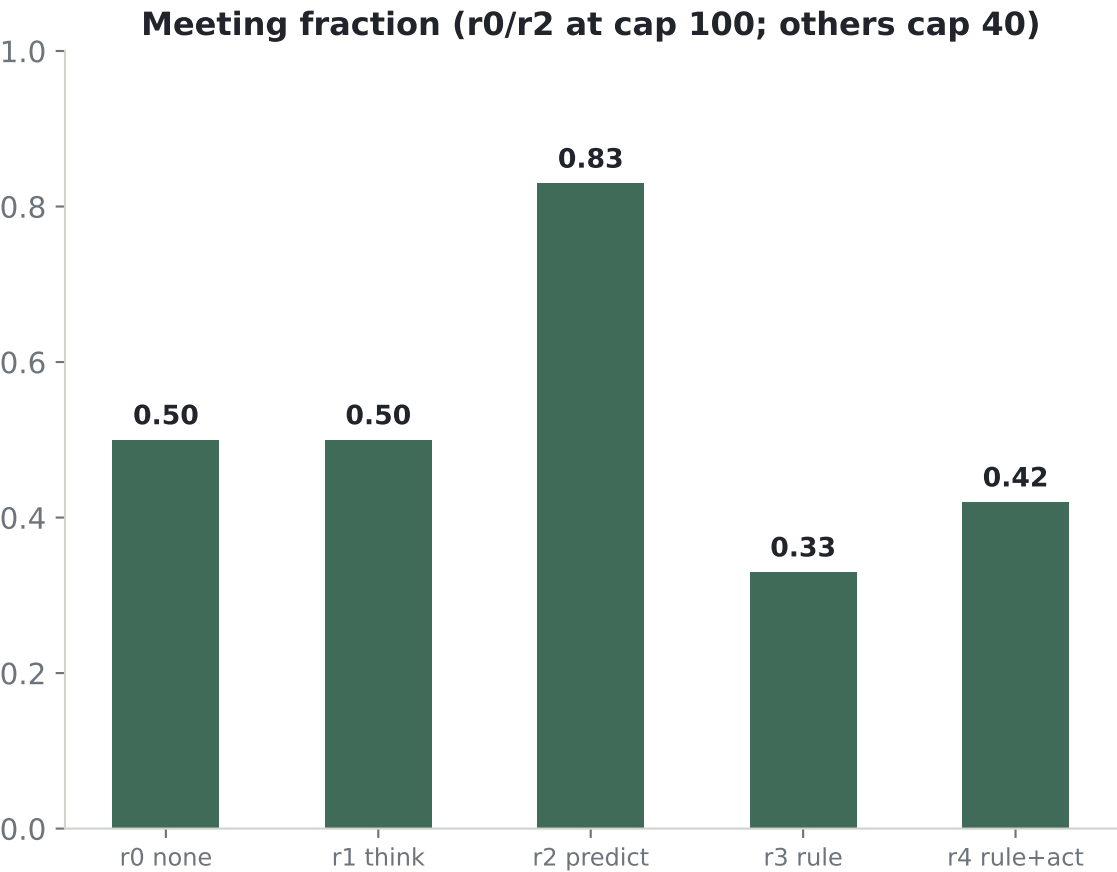
source: 3_interventions/seed_matrix/seed_matrix.json (dose-3 attribution + lexical cells, n=6 replayed streams each)

"Predict your partner" helps; "identify the rule" backfires

"Silently predict what the other player will say next" doubles category mass and lifts meeting to 0.83 (vs 0.50 baseline at cap 100) — and the effect GROWS with horizon (0.42 at cap 40).

"Identify the rule behind the other player's choices" collapses category mass to 0.008 and TRIPLES self-family mass — explicit meta-reasoning about "the rule" redirects attention to the rule itself.

Survivorship caveat: turn-8 probes condition on games still alive at t8.

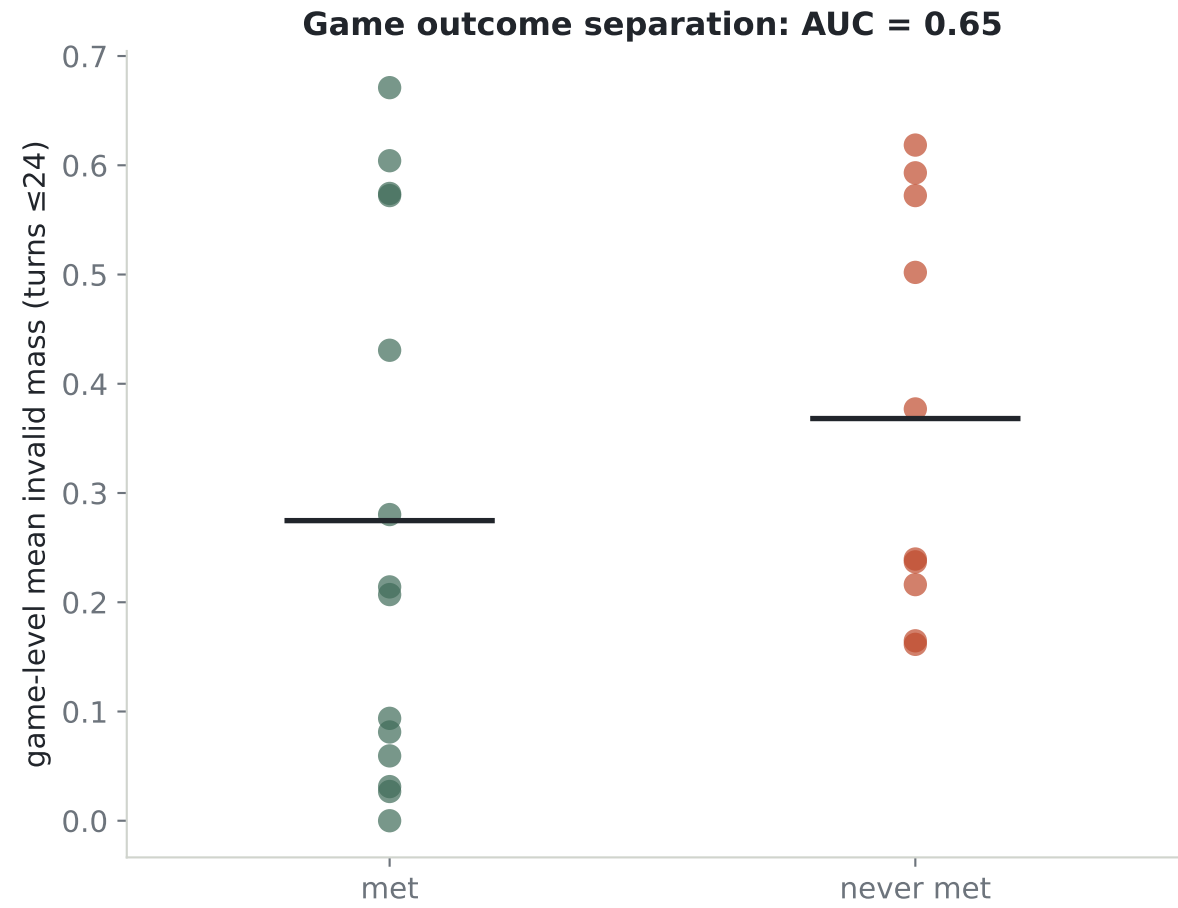
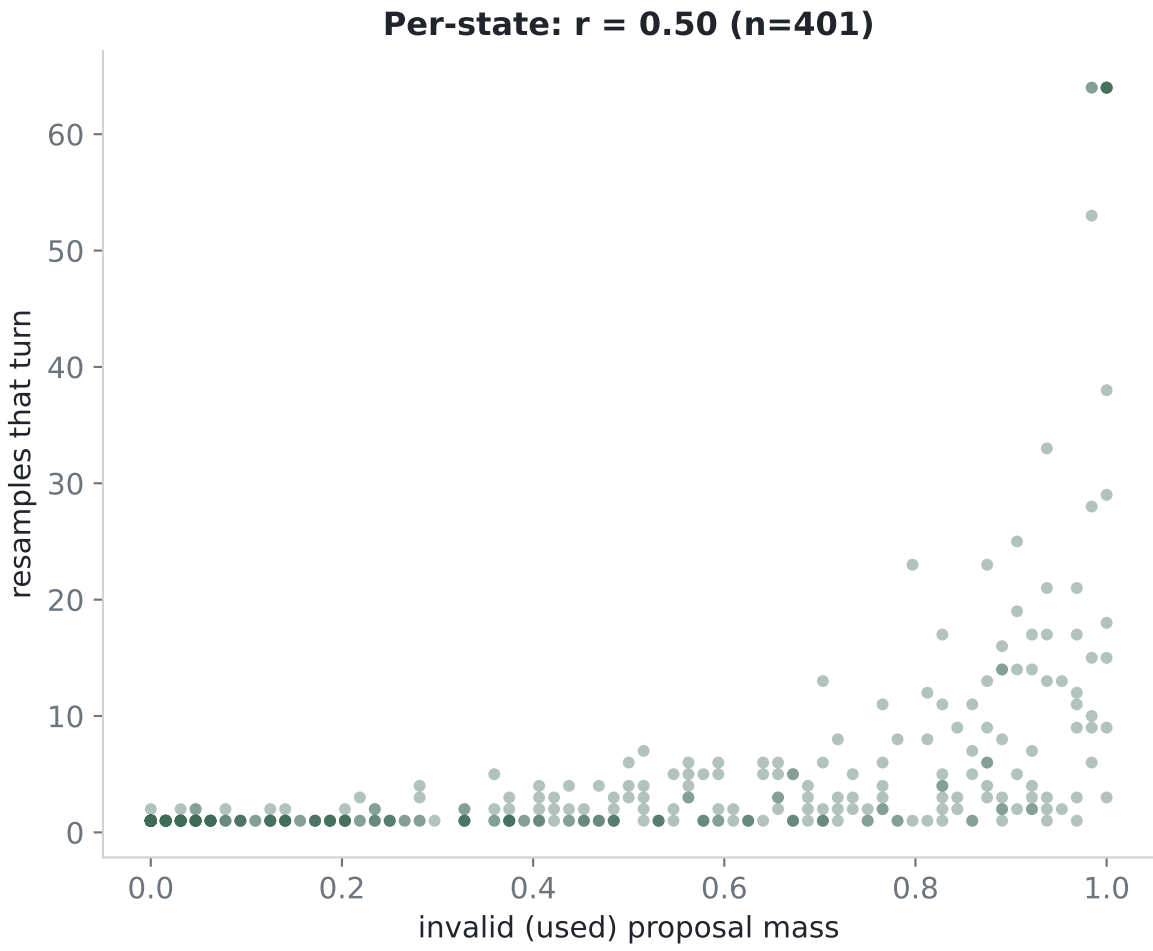


source: pod intervention_ladder runs (cap-40 all rungs; cap-100 rerun of r0/r2) · prompt additions to A only

Collapse predicts resample burden and coordination failure

Invalid proposal mass correlates with the resample burden the handler pays ($r = 0.50$) and separates game outcomes ($AUC = 0.65$; means 0.27 met vs 0.37 no-meet).

MC samples within a game are not independent — treat per-state r as descriptive; game-level AUC is the honest unit.



Caveats, detector debt, and what is running now

Measurement caveats

Turn-8 / per-turn probes condition on games surviving to the probe (survivorship); a naive pooled probe once scored 0.99 from this bias alone.

The original per-turn logit telemetry read a formatting-token position (entropy ≈ 0.1) — deprecated; all collapse numbers here are MC proposal profiling.

Used-word mass partly scales mechanically with game length; self-family and category masses carry the load-bearing comparisons.

MC proposals within a game are not independent; game-level statistics are the honest unit.

Detector debt (discovered via Llama-70B, 2026-08-02)

All onset/self-family metrics are 4-PREFIX-only. Llama-70B loops on a SUFFIX/RHYME dimension the detector cannot see

(cursed→blamed→flamed→hamed→mamed) — Qwen32 conclusions are prefix-complete, but cross-model 'immunity' claims need a dimension-general lexical-family detector (prefix / suffix / embedding cluster) re-run over all transcripts.

In flight

Qwen-72B cap-100 games (pod wave6) · Llama-70B strict-masking + raw-proposal telemetry (pod wave7) — the direct test of whether Llama's repeat-locks are resample-exhaustion artifacts or a genuine Qwen-style proposal collapse.